

BEE 4750 Lab 2: Monte Carlo

Name:

ID:

Due Date

Thursday, 10/02/25, 9:00pm

Setup

The following code should go at the top of most Julia scripts; it will load the local package environment and install any needed packages. You will see this often and shouldn't need to touch it.

```
import Pkg
Pkg.activate(".")
Pkg.instantiate()
```

```
Activating project at `c:\Users\elrae\BEE4750\lab-2-ellaharrod-1`
```

```
using Random # random number generation
using Distributions # probability distributions and interface
using Statistics # basic statistical functions, including mean
using Plots # plotting
```

Overview

In this lab, we will conduct a Monte Carlo experiment and explore how sensitive Monte Carlo estimates are to the underlying probability distribution(s).

You should always start any computing with random numbers by setting a “seed,” which controls the sequence of numbers which are generated (since these are not *really* random, just “pseudorandom”). In Julia, we do this with the `Random.seed!()` function.

```
Random.seed!(1)
```

`TaskLocalRNG()`

It doesn't matter what seed you set, though different seeds might result in slightly different values. But setting a seed means every time your notebook is run, the answer will be the same.

Seeds and Reproducing Solutions

If you don't re-run your code in the same order or if you re-run the same cell repeatedly, you will not get the same solution. If you're working on a specific problem, you might want to re-use `Random.seed()` near any block of code you want to re-evaluate repeatedly.

Probability Distributions and Julia

Julia provides a common interface for probability distributions with the [Distributions.jl package](#). The basic workflow for sampling from a distribution is:

1. Set up the distribution. The specific syntax depends on the distribution and what parameters are required, but the general call is the similar. For a normal distribution or a uniform distribution, the syntax is

```
# you don't have to name this "normal_distribution"
#  is the mean and  is the standard deviation
normal_distribution = Normal( , )
# a is the upper bound and b is the lower bound; these can be set to +Inf or -Inf for an
uniform_distribution = Uniform(a, b)
```

There are lots of both [univariate](#) and [multivariate](#) distributions, as well as the ability to create your own, but we won't do anything too exotic here.

2. Draw samples. This uses the `rand()` command (which, when used without a distribution, just samples uniformly from the interval $[0, 1]$.) For example, to sample from our normal distribution above:

```
# draw n samples
rand(normal_distribution, n)
```

Putting this together, let's say that we wanted to simulate 100 six-sided dice rolls. We could use a [Discrete Uniform distribution](#).

```
dice_dist = DiscreteUniform(1, 6) # can generate any integer between 1 and 6
dice_rolls = rand(dice_dist, 100) # simulate rolls
```

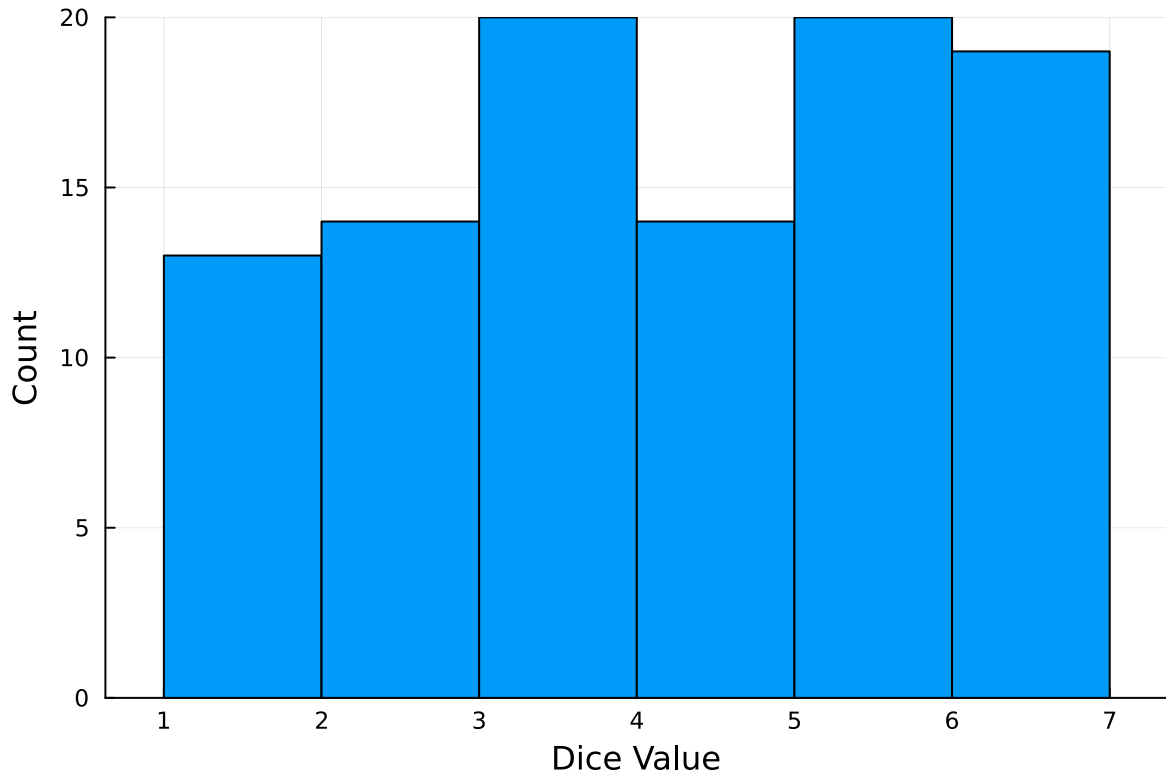
100-element Vector{Int64}:

```
1
3
5
4
6
2
5
5
5
2

3
6
5
5
6
3
6
6
6
```

And then we can plot a histogram of these rolls:

```
histogram(dice_rolls, legend=:false, bins=6)
ylabel!("Count")
xlabel!("Dice Value")
```



Instructions

Remember to:

- Evaluate all of your code cells, in order (using a `Run All` command). This will make sure all output is visible and that the code cells were evaluated in the correct order.
- Tag each of the problems when you submit to Gradescope; a 10% penalty will be deducted if this is not done.

Problem

A common engineering problem is to quantify flood risk, which is typically computed by propagating a flood hazard distribution through a *depth-damage function* relating flood depths to economic damages. A reasonable depth-damage function for a house without a basement is a bounded logistic function,

$$d(h) = \mathbb{1}_{h>0} \frac{L}{1 + \exp(-k(h - h_0))},$$

where d is the damage as a percent of total structure value, h is the water depth (relative to the house's ground floor elevation) in m, $\mathbb{1}_{x>0}$ is the indicator function, L is the maximum loss in USD, k is the slope of the depth-damage relationship, and h_0 is the inflection point. We'll assume $L = \$200,000$, $k = 0.8$, and $h_0 = 3$.

For this problem, suppose that we have two different probability distributions characterizing annual maxima flood depths:

1. $h \sim \text{LogNormal}(1.2, 0.3)$;
2. $h \sim \text{GeneralizedExtremeValue}(3, 0.9, -0.15)$.

Plot histograms of samples from these distributions and conduct a Monte Carlo experiment for annual maximum flood damages using each. Answer the following questions:

- What are the Monte Carlo estimates of the expected value and the 99% quantile for the annual maximum damage the structure would suffer for each of these flood hazard distributions?
- How did you decide on your sample size?
- Why do you think the estimates differed or did not differ (you can also plot the depth-damage function to compare with the samples)?
- How might you decide (based on getting additional information, if needed) which distribution is “better”?

```
import Pkg; Pkg.add("StatsBase")

# Depth-damage function
function damage(h; L=200_000, k=0.8, h0=3.0)
    if h > 0
        return L / (1 + exp(-k * (h - h0)))
    else
        return 0.0
    end
end

# Monte Carlo simulation
function monte_carlo_dam(dist; N=75000)
    hs = rand(dist, N)
    ds = [damage(h) for h in hs]
    = mean(ds)
    q99 = quantile(ds, 0.99)
    return , q99, ds
end
```

```

# Define the distribution
dist1 = LogNormal(1.2, 0.3)

# Run simulations
1, q99_1, ds1 = monte_carlo_dam(dist1)

println("LogNormal(1.2, 0.3):")
println(" Expected damage = \$$ (round(1, digits=2))")
println(" 99% quantile damage = \$$ (round(q99_1, digits=2))")

# Plot
histogram(ds1, bins=50, alpha=0.5, color=:teal,
xlabel="Damage (in Dollars)", ylabel="Frequency", title="LogNormal Damage Histograms", label=

```

```

    Resolving package versions...
    No Changes to `C:\Users\elrae\BEE4750\lab-2-ellaharrod-1\Project.toml`
    No Changes to `C:\Users\elrae\BEE4750\lab-2-ellaharrod-1\Manifest.toml`

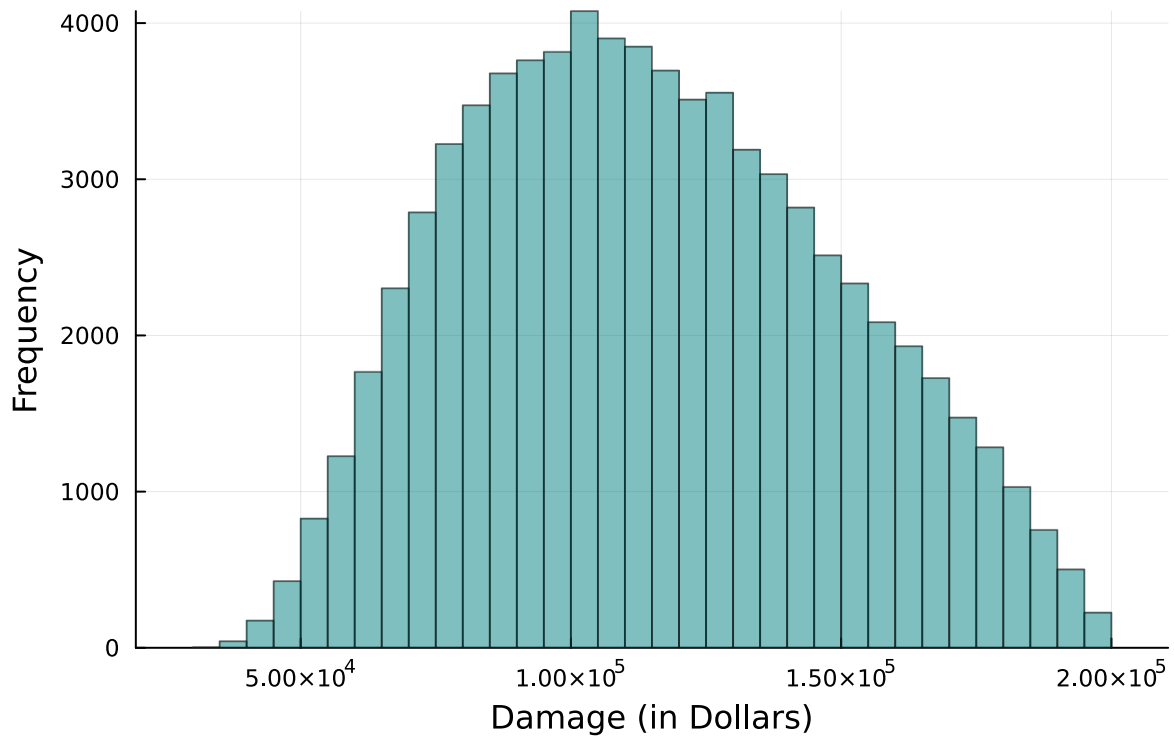
```

```

LogNormal(1.2, 0.3):
Expected damage = $115051.54
99% quantile damage = $189817.74

```

LogNormal Damage Histograms



```
import Pkg; Pkg.add("StatsBase")

# Depth-damage function
function damage(h; L=200_000, k=0.8, h0=3.0)
    if h > 0
        return L / (1 + exp(-k * (h - h0)))
    else
        return 0.0
    end
end

# Monte Carlo simulation
function monte_carlo_damage(dist; N=75000)
    hs = rand(dist, N)
    ds = [damage(h) for h in hs]
    = mean(ds)
    q99 = quantile(ds, 0.99) # quantile
    return , q99, ds
end
```

```

# Define the distributions
dist2 = GeneralizedExtremeValue(3.0, 0.9, -0.15)

# Run simulations
2, q99_2, ds2 = monte_carlo_damage(dist2)

println("\nGEV(3, 0.9, -0.15):")
println(" Expected Damage = \$$ (round(2, digits=2))")
println(" 99% quantile damage = \$$ (round(q99_2, digits=2))")

# Plot histograms
histogram(ds2, bins=50, alpha=0.5, color=:orange, xlabel="Damage (in Dollars)", ylabel="Frequency")

```

```

Resolving package versions...
No Changes to `C:\Users\elrae\BEE4750\lab-2-ellaharrod-1\Project.toml`
No Changes to `C:\Users\elrae\BEE4750\lab-2-ellaharrod-1\Manifest.toml`

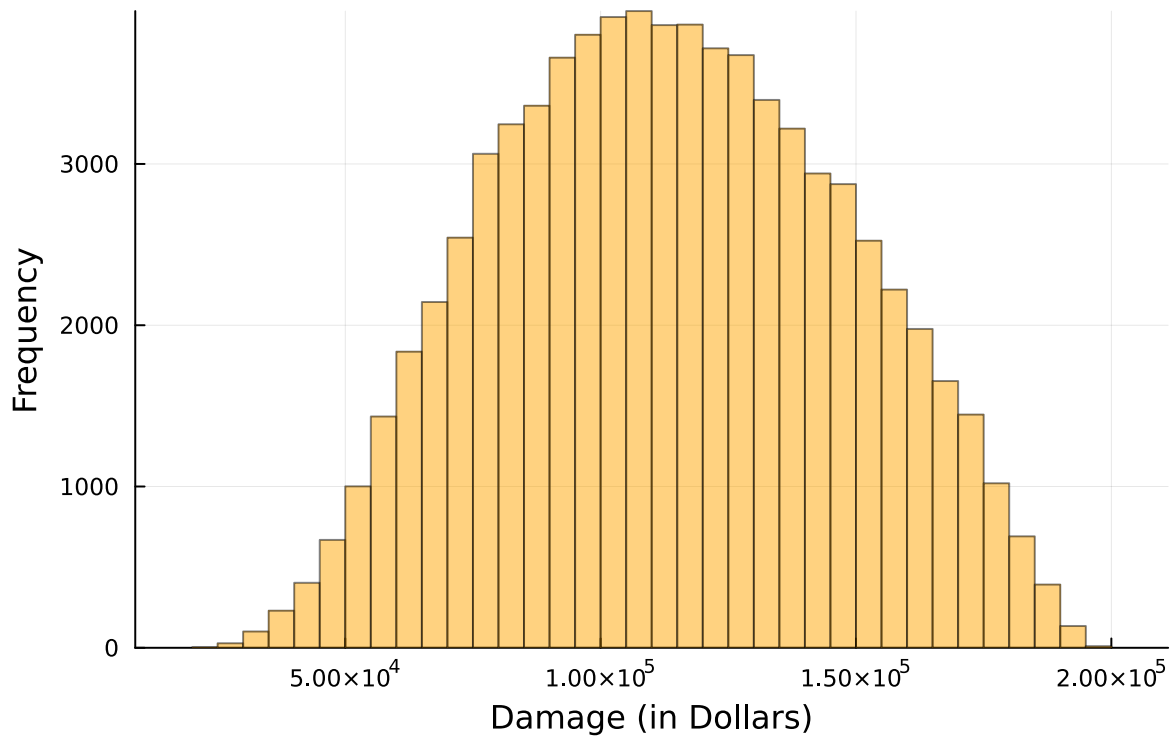
```

```

GEV(3, 0.9, -0.15):
Expected Damage = $113510.31
99% quantile damage = $183302.4

```


GEV Damage Histograms



References

Put any consulted sources here, including classmates you worked with/who helped you.